6.6.4 Blood Pressure

Blood pressure (BP) is the lateral force on the walls of artery by pulsing blood under pressure from the heart. The hearts contraction forces blood under high pressure into the aorta. The peak of maximum pressure when ejection occurs is the systolic blood pressure. When the ventricles relax, the blood remaining in the arteries exerts a minimum a diastolic pressure.Diastolic pressure is the minimal pressure exerted against the arterial walls at all times.The standard unit for measuring blood pressure is millimeters of mercury (mmHg). The BP is recorded with the systolic reading before diastolic.

E.g. 120/80 mmHg. 120 is systolic pressure and 80 is diastolic pressure. The difference between systolic and diastolic pressure is pulse pressure.

Physiology of BP

BP reflects inter relationship of cardiac output, peripheral vascular resistance. Cardiac output is the volume of blood pumped by the heart (stroke volume) in one minute.

- Cardiac output = heart rate × stroke volume
- The BP depends on the cardiac output and peripheral vascular resistance (R).
- BP = cardiac output × R

Peripheral vascular resistance is the resistance to blood flow determined by the vascular wall and diameter of blood vessels when the diameter is less, the vascular resistance to blood flow is increased.

Blood volume the volume of blood circulating with in vascular system, affects BP. For adults, normal circulating blood volume is 5000 ml. If volume increases, BP elevates. E.g. rapid uncontrolled intravenous fluid. When circulating volume falls, blood pressure falls. E.g. hemorrhage, dehydration.

Elasticity: normally the arterial walls are elastic and easily distensible. In diseases such as arteriosclerosis, lose their elasticity and cannot stretch wall. When the blood is forced into blood vessels, due to the rigid blood walls, the systolic pressure rises.

Under normal conditions autonomic reflexes stabilize an individual's blood pressure from the lying to sitting/standing positions.

Factors influencing variations in BP

- **Age:** Normal BP levels vary throughout life big children have higher BP than smaller children of same age.
- **Stress:** Anxiety, fear, pain and emotional stress result in increase in heart rate resulting in increasing in BP
- **Race:** Certain races are more prone for high BP genetically and environmentally.
- **Medication:** some medications can affect BP directly or indirectly.
- **Diurnal variation:** BP levels vary over the course of a day. BP is lowest in the

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